

Learning Criticality with a Temperature-Conditioned GFlowNet

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Abstract

A temperature-conditioned GFlowNet is trained to sample the periodic two-dimensional Ising model and expose criticality through its learned normalization. Exact L=4 enumeration and seeded Metropolis chains through L=12 gate every result. Fixed-temperature empirical KL is below 0.016 nats; a conditioned model reaches L=4 exact KL below 0.0033 and L=8 observable errors below 4.35%. Calibrating the learned-log-Z route by its 6.5% L=4 known-truth miss gives $T_c = 2.34 \pm 0.15$. Susceptibility extrapolation gives 2.274 and the Binder mean 2.246, an observable range of about 2.25 to 2.27 around exact 2.2692; their opposite-sign biases are not averaged. The study identifies nonphysical boundary curvature, finite-size bias, and regimes where MCMC remains preferable.

Keywords: GFlowNet; Ising model; trajectory balance; partition function; critical temperature; finite-size scaling

1. Introduction

The two-dimensional zero-field Ising model is a compact test of generative sampling near a phase transition. Below criticality it has two symmetry-related ordered modes; around criticality it develops long correlations; and its partition function is normally the hard normalization in the Boltzmann law. This study asks whether a small generative flow network (GFlowNet), trained on one Mac CPU, can reproduce the distribution and predict critical temperature from its learned structure rather than merely copying a known answer.

The study has three safeguards against impressive-looking but incorrect results. First, all 65,536 states of the L=4 lattice are exactly enumerated. Second, seeded Metropolis-Hastings sampling provides an independent baseline at every tested temperature. Third, milestone validators enforce numerical thresholds and write immutable JSON provenance. The exact thermodynamic-limit value, $T_c = 2/\ln(1+\sqrt{2}) = 2.269185$, is reserved for reference and is never supplied to training [4].

The central contribution is two independent routes. Calibrating the learned-log-Z route by its 6.5% L=4 known-truth miss gives $T_c(\log Z) = 2.34 \pm 0.15$. The observable route is kept as two estimates: susceptibility extrapolation 2.274 and Binder mean 2.246, a range of about 2.25 to 2.27. Their finite-size biases have opposite signs, so averaging them overstates precision.

The computation and original report build ran in one automated session of roughly 90 minutes wall-clock.

2. Model and Methods

2.1 Ising target and observables

Spins s_i are -1 or +1 on an L by L periodic square lattice with nearest-neighbor coupling $J=1$ and zero field. Counting every bond once, the Hamiltonian and Boltzmann law are

$$E(x) = -\sum_{\langle ij \rangle} s_i s_j, \quad p_T(x) = \exp[-E(x)/T] / Z(T).$$

With $N=L^2$ and $M=\sum_i s_i$, the analysis records energy per site, absolute magnetization, connected absolute-magnetization susceptibility, specific heat, and Binder cumulant:

$$\chi = \beta [\langle M^2 \rangle - \langle M \rangle^2] / N, \quad U_4 = 1 - \langle M^4 \rangle / (3 \langle M^2 \rangle^2).$$

The absolute-magnetization convention prevents finite-volume tunneling between the two ordered signs from appearing as a spurious susceptibility. Exact L=4 values use stable log-sum-exp evaluation. Metropolis uses checkerboard single-spin sweeps with acceptance $\min[1, \exp(-\Delta E/T)]$, mixed starts, burn-in, thinning, and explicit independent seeds.

2.2 Raster-order GFlowNet

The source is an unassigned lattice. At raster step t , the forward policy assigns -1 or +1 to site t . The terminal reward is $R_T(x) = \exp[-E(x)/T]$. Raster order creates one trajectory per terminal and one parent per non-source state, so backward transition probabilities equal one. General trajectory balance [2] reduces to

$$\Delta(x, T) = \log Z(T) + \sum_t \log P_T(s_{t+1} | s_t, T) + E(x)/T.$$

Training minimizes the mean squared residual. When balance holds for every complete trajectory, policy normalization forces $P_T(x|T) = R_T(x)/Z(T)$. Thus $\log Z$ is learned rather than supplied by the oracle. This is the normalized discrete-model use of GFlowNets described in Refs. [1-3].

2.3 Temperature conditioning and symmetry

Fixed-temperature L=4 models use two 128-unit hidden layers. The conditioned models take $\beta=1/T$ and use a small separate network for $\log Z(\beta)$. A masked autoregressive MLP returns every site conditional in one training pass while blocking present and future spins. The L=8 policy uses two 256-unit hidden layers, the CPU ceiling, plus spin and beta-times-spin features.

Zero-field symmetry is imposed exactly by replacing a raw logit $g(s, \beta)$ with $[g(s, \beta) - g(-s, \beta)]/2$. Consequently $q(x|\beta) = q(-x|\beta)$, the first spin is unbiased, and neither ordered sign is structurally favored. Training batches combine uniform, current-policy, noisy ordered, and block-domain configurations, paired with global flips. This off-policy mixture preserves full support while exposing low-energy and domain-wall sectors.

3. Validation Results

3.1 Physics core and fixed-temperature models

The final test suite contains 20 passing checks covering energy signs, wrapped bonds, exact enumeration, seeded sampling, autoregressive normalization and causality, spin symmetry, and numerical peak estimators. Table 1 shows the worst L=4 Metropolis deviations over the M1 temperature grid.

Metric	Worst error	Gate
Energy/site	0.1003%	1%
m	0.1019%	1%
Susceptibility	2.7492%	5%
Specific heat	1.2070%	5%

Table 1. Exact-oracle validation. Source: M1 JSON.

For M2, two million seeded multinomial draws from exactly enumerated model probabilities give $KL(\text{exact}||\text{empirical}) = 0.014762$ at $T=3$ and 0.015377 at $T=2$. For this normalized autoregressive model the draws are distributionally equivalent to sequential rollout terminals, but they were not rollouts. Exact enumerated model KL values are 0.000511 and 0.002200, and learned log Z errors are 0.0055% and 0.5473%. The histogram KL uses a disclosed Jeffreys half-count so physically nonzero rare states do not receive zero probability.

3.2 One conditioned model

The single conditioned L=4 model reaches exact KL 0.003255, 0.001690, and 0.000308 at $T=1.8$, 2.269185, and 3.0. Its maximum log Z value error over a 16-point grid is 0.1053%. Table 2 compares 200,000 L=8 states from true sequential rollouts with 256,000 fresh Metropolis states at each temperature. M4 likewise used true sequential rollouts for its generated-observable curves.

T	E error	m error	chi error
1.8	0.0034%	0.0281%	4.3499%
2.269	0.2359%	0.4480%	2.7736%
3.0	0.2462%	0.3822%	0.6978%

Table 2. Conditioned L=8 relative errors. Source: M3 JSON.

At $T=1.8$ the generated fraction $P(m>0)=0.498825$, meeting the 0.50 ± 0.05 mode criterion. At higher T the raw positive fraction falls slightly because exactly zero magnetization becomes more common; conditional on nonzero magnetization the signs remain balanced.

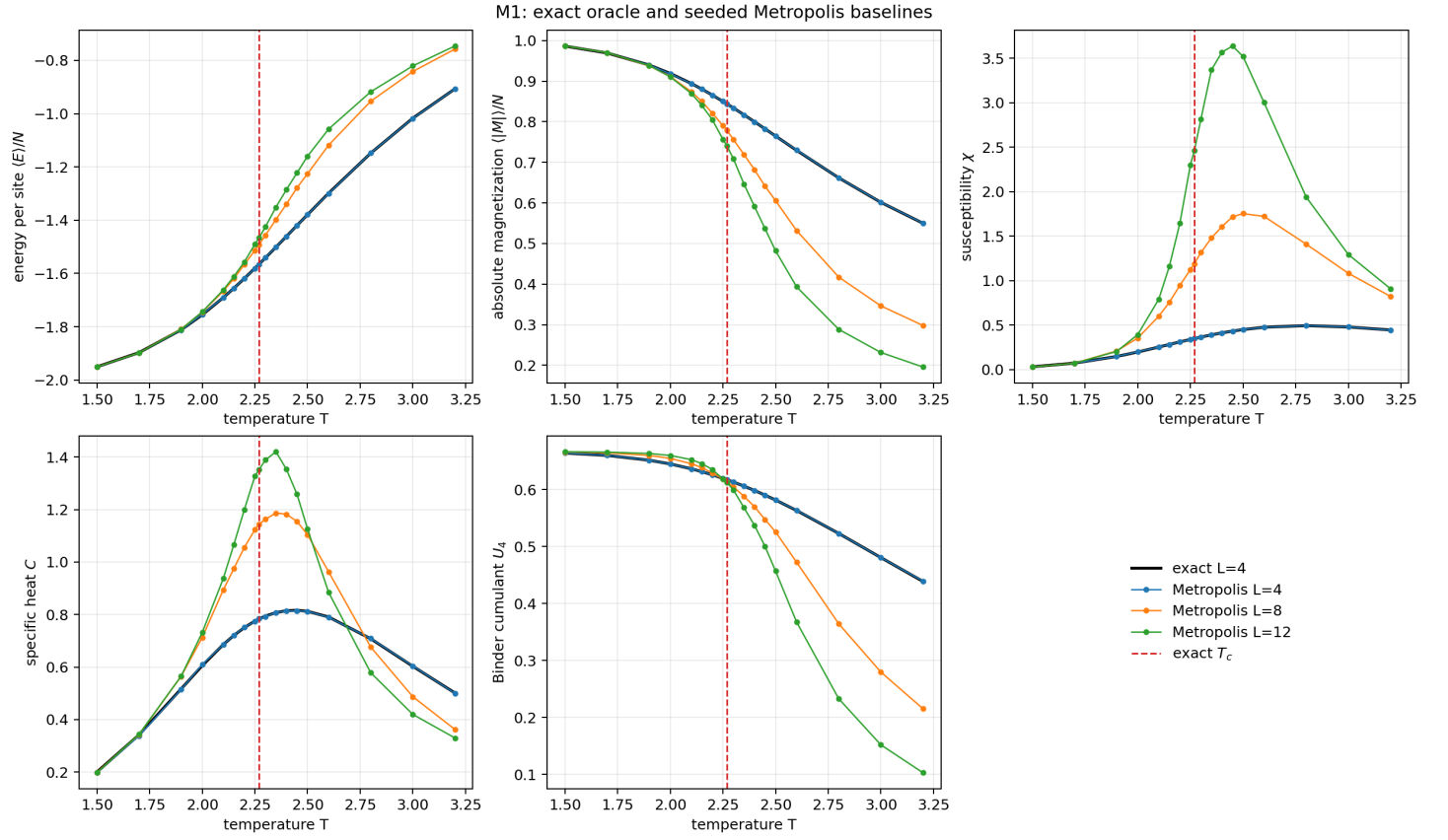


Figure 1. Exact $L=4$ observables and seeded Metropolis baselines at $L=4, 8$, and 12 . Every temperature-axis panel marks the exact thermodynamic-limit critical temperature. Source: M1.

Artifact: results/m1_observables_20260818T002648-0400.png

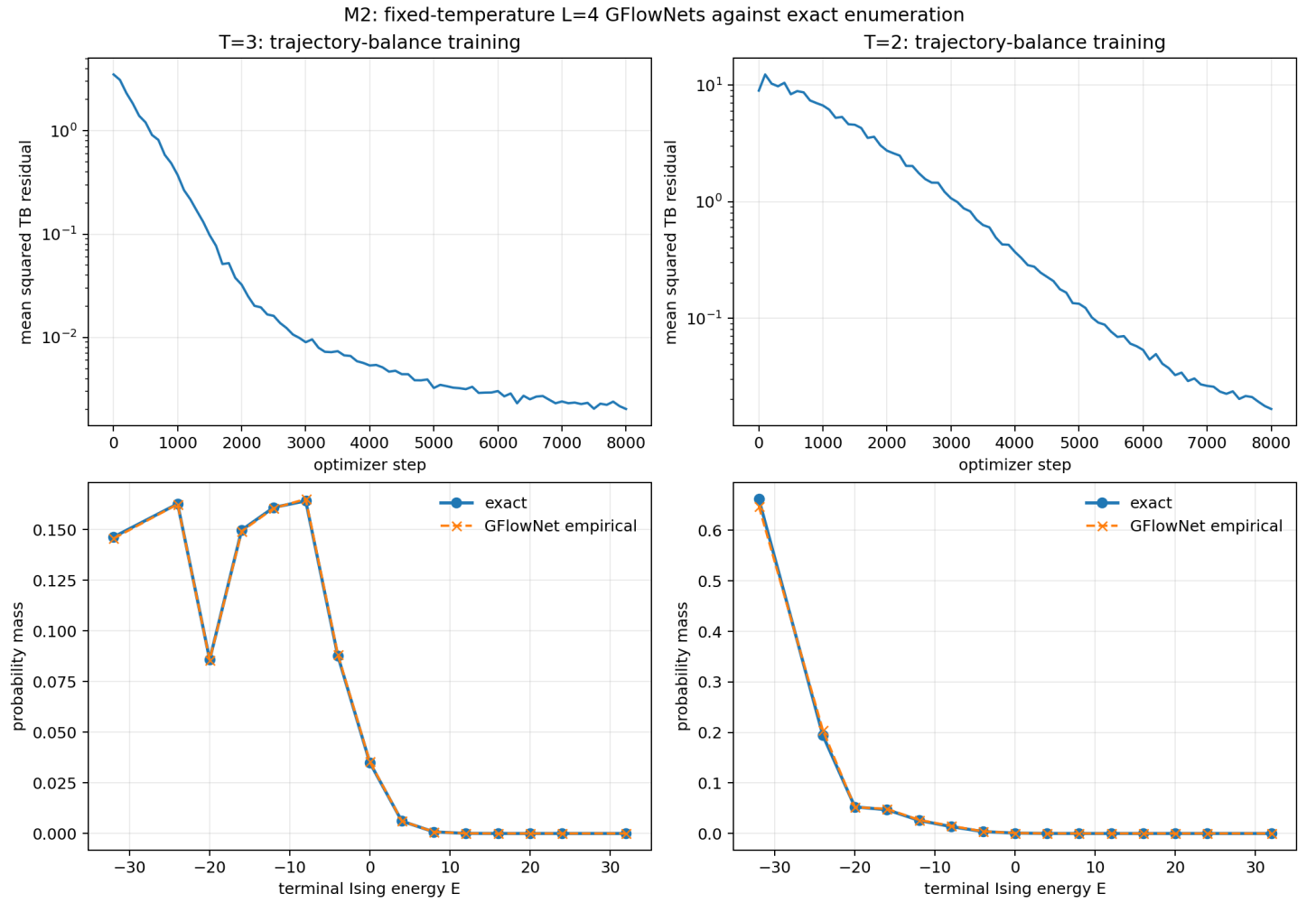


Figure 2. Fixed-temperature trajectory-balance convergence and energy-level probability mass. Empirical model mass overlays exact enumeration at both temperatures. Source: M2.

Artifact: results/m2_fixed_temperature_20260818T012603-0400.png

4. Predicting Critical Temperature

4.1 Learned partition-function route

The partition function supplies thermodynamics before terminal sampling. In beta coordinates, $U = -d \log Z / d \beta$ and $c = \beta^2 d^2 \log Z / d \beta^2 / N$. M4 evaluates $\log Z$ on 256 beta points, fits a degree-10 Chebyshev polynomial, differentiates it analytically, and searches for the interior heat-capacity maximum.

Direct exact $L=4$ observables peak at $T=2.438950$, while the exact-log- Z pipeline returns 2.439257, a 0.0126% location error. This validates differentiation on exact data, but not the learned $\log Z$ itself. The learned $L=4$ peak is 2.281360, a 6.5% known-truth miss. Applying that empirical relative calibration error to the raw $L=8$ central peak gives

$$T_c^{(\log Z)} = 2.34 \pm 0.15.$$

This is an empirical calibration bar, not a statistical confidence interval. The differentiation pipeline was validated on exact data, but the learned $\log Z$ itself carries this calibration error. The central value is in the predeclared [2.1,2.5] window. The differentiated learned curves also become negative near $T=1.5$; a canonical Ising heat capacity cannot be negative, so this is nonphysical boundary curvature, not a physical discovery.

4.2 Generated-observable route

One hundred thousand new GFlowNet samples at each of 20 temperatures supply $L=4$ and $L=8$ observables; 144,000 new Metropolis samples per temperature supply $L=12$. Local quadratic fits place susceptibility maxima at 2.812028, 2.552567, and 2.447225 for $L=4$, 8, and 12. Fitting $T_{\text{chi}}(L) = T_c + a/L$ gives

$$T_c^{(\text{chi})} = 2.273526, \quad R^2 = 0.99825.$$

As a methodological control, the identical procedure on the existing M1 Metropolis-only $L=4$, 8, and 12 curves gives peaks 2.830413, 2.550983, and 2.446206, followed by

$$T_c^{(\text{chi}, M1 \text{ MCMC})} = 2.259472, \quad R^2 = 0.99941.$$

No new sampling was performed. The coarser M1 grid makes this a method control rather than a precision result: switching from a five-point to a three-point local fit shifts its intercept by about 0.008. Linear Binder crossings are 2.242534 and 2.249908, with mean 2.246221. Thus the observable estimates are susceptibility 2.274 and Binder 2.246, a range of about 2.25 to 2.27. The first is 0.19% high and the second 1.01% low; their biases have opposite signs, so averaging them overstates precision. No observable consensus is reported.

4.3 Trajectory signatures

The $L=8$ mean Bernoulli action entropy rises from 0.04587 nats at $T=1.5$ through 0.28136 at exact criticality to 0.56129 at $T=3.2$. This smooth increase reflects the change from ordered, predictable later actions to disordered choices. At exact criticality $P(m>0|m!=0)=0.50014$. These are exploratory structural signatures; no third critical temperature is extracted from them.

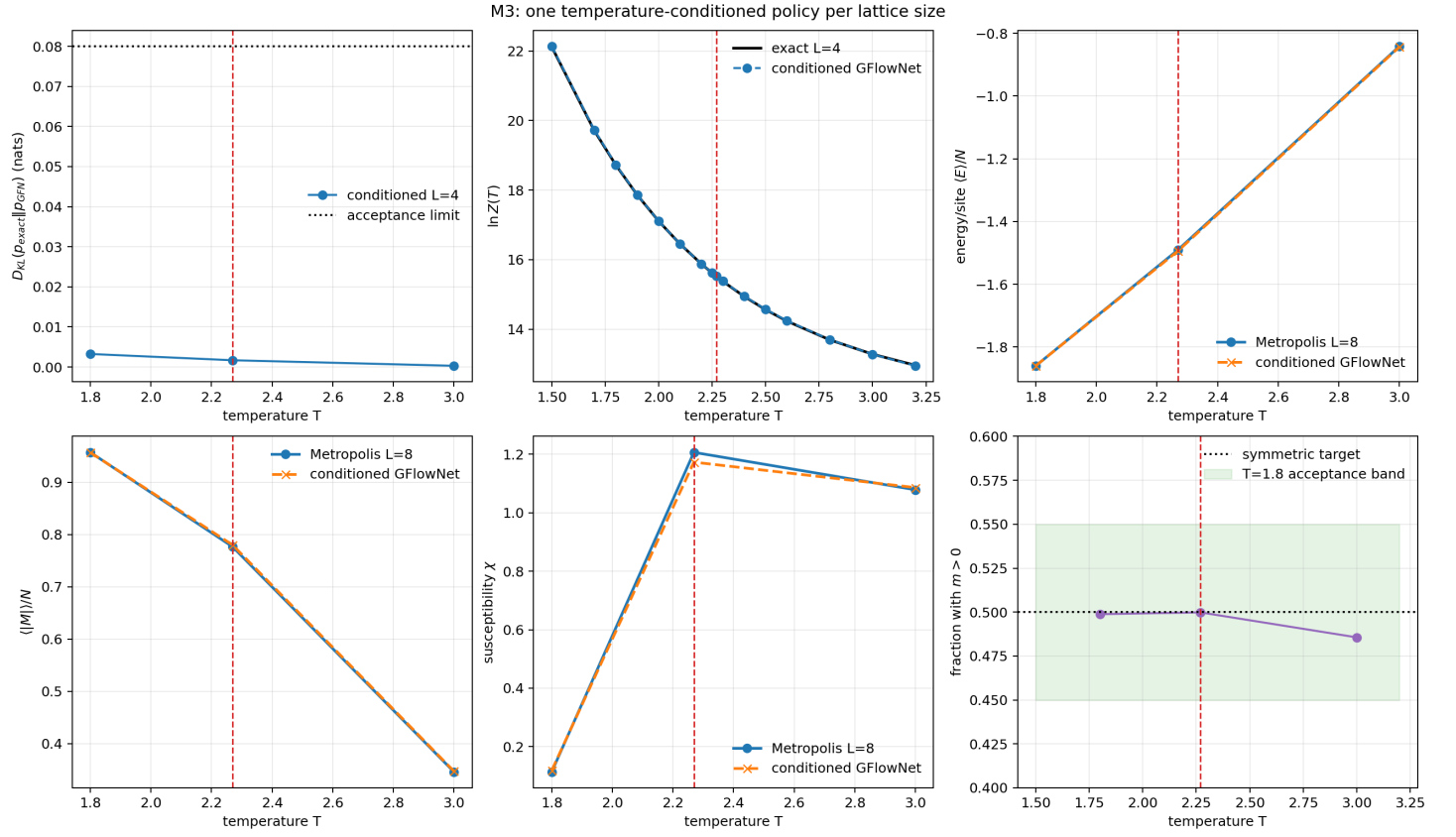


Figure 3. Validation of one temperature-conditioned policy per lattice size: exact L=4 KL and log Z, L=8 observables against Metropolis, and terminal-mode balance. Source: M3.

Artifact: results/m3_conditioned_summary_20260818T015650-0400.png

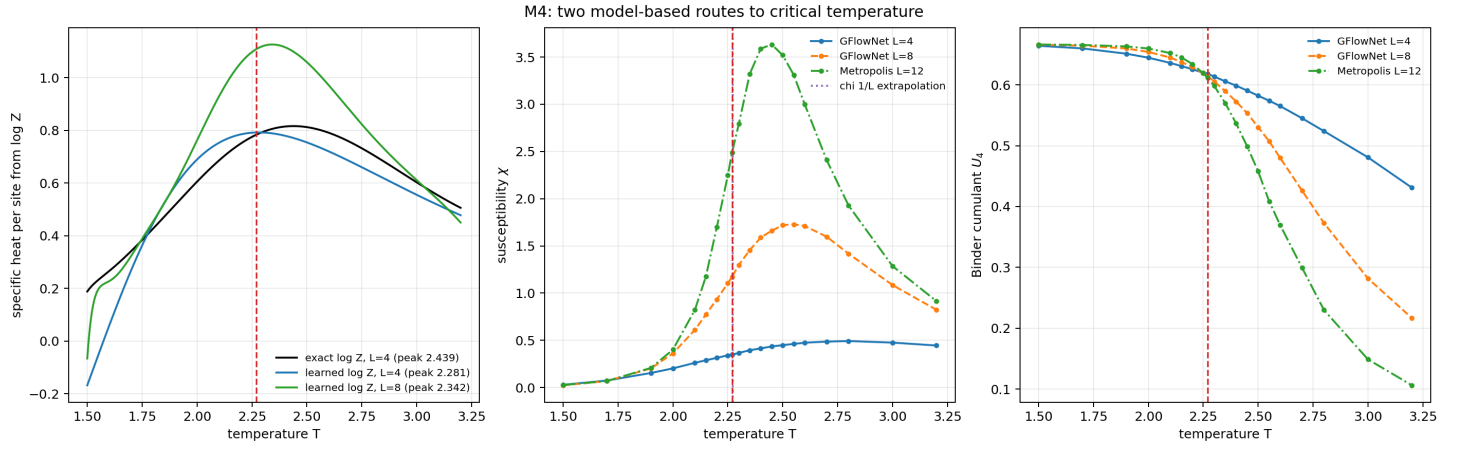


Figure 4. The required M4 summary: heat capacity from learned $\log Z$, susceptibility, and Binder cumulant, all with the exact critical-temperature line. Source: M4.

Artifact: results/m4_tc_summary_20260818T021212-0400.png

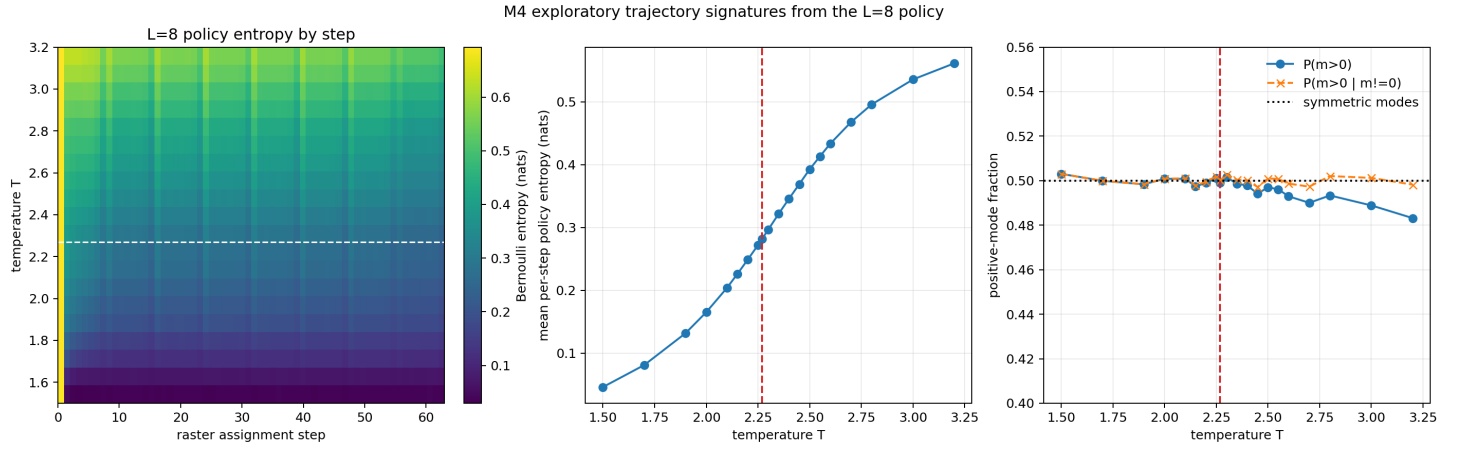


Figure 5. Exploratory $L=8$ policy entropy by raster step and temperature, mean entropy, and positive-mode balance. Source: M4.

Artifact: results/m4_trajectory_signatures_20260818T021212-0400.png

5. Discussion

The strongest evidence is the agreement of several independent objects: exact $L=4$ distributions, learned $\log Z$ values, $L=8$ generated observables, $L=12$ Metropolis trends, and separate finite-size critical estimators. The GFlowNet does more than match a mean. Its normalized autoregressive distribution covers both ordered signs and its learned normalization contains enough curvature to locate an interior heat-capacity peak.

Finite size remains the dominant physical limitation. $L=4$ and $L=8$ are far from the thermodynamic limit; their rounded susceptibility peaks shift substantially. The high R-squared of the three-point $1/L$ line only describes those three points and is not asymptotic evidence. Repeated training seeds and bootstrap confidence intervals are also absent, so displayed digits identify this reproducible run rather than universal precision.

Derivative sensitivity is the principal model limitation. Sub-percent errors in $\log Z$ values can become visible curvature errors after two derivatives, which is why the $L=4$ known-truth miss calibrates the $L=8$ result. A future model should parameterize $\log Z$ as a convex function of β , because its second derivative is $\text{Var}(E)$ and must be nonnegative. Larger locality-aware policies and repeated-seed uncertainty estimates are natural follow-ups.

MCMC still wins for a one-off trusted estimate on a modest lattice: it needs no neural training, its transition rule is transparent, and it reached $L=12$ under the same CPU budget. The GFlowNet wins after amortization when many independent samples across temperature are required, when explicit two-mode generation matters, or when learned $\log Z$ is itself the object of study. The methods are complementary rather than interchangeable.

6. Conclusion

A compact CPU GFlowNet learned the periodic 2D Ising distribution to exact-oracle accuracy at $L=4$, generalized across temperature, and reproduced $L=8$ Metropolis observables within the declared gates. The calibrated learned partition function gives $T_c = 2.34 \pm 0.15$. Generated observables give separate susceptibility and Binder estimates of 2.274 and 2.246, spanning about 2.25 to 2.27 around exact 2.2692. Their opposite-sign biases are not averaged into a falsely precise consensus. The $\log Z$ route remains the distinctively model-based result and reveals the numerical fragility of thermodynamic derivatives.

Artifact Provenance

Every quantitative claim above is regenerated by an acceptance-gated validator. Checkpoints are SHA-256 hashed in their JSON records. The report and this PDF introduce no uncomputed table entries.

ID	Role	Primary artifact
M1	Exact + MCMC	results/m1_metrics_20260818T002648-0400.json
M2	Fixed-T GFN	results/m2_metrics_20260818T012603-0400.json
M3	Conditioned GFN	results/m3_metrics_20260818T015650-0400.json
M4	Criticality	results/m4_metrics_20260818T021212-0400.json
M5	Report audit	results/m5_metrics_20260818T030253-0400.json

Table 3. Machine-readable provenance map.

References

- [1] Y. Bengio et al., “GFlowNet Foundations,” arXiv:2111.09266 (2021).
- [2] N. Malkin et al., “Trajectory Balance: Improved Credit Assignment in GFlowNets,” arXiv:2201.13259 (2022).
- [3] D. Zhang et al., “Generative Flow Networks for Discrete Probabilistic Modeling,” arXiv:2202.01361 (2022).
- [4] L. Onsager, “Crystal Statistics. I. A Two-Dimensional Model with an Order-Disorder Transition,” Physical Review 65, 117-149 (1944).